WORKED SOLUTION

STEP III 1995, Question 11

Mechanics • Relative Velocity

STEP III 1995, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q11

The key idea

Work in the ship's frame. Relative to the ship, the plane has to fly straight out along the north-west line and straight back again. The air carries the plane, so in this frame the plane's velocity is its airspeed vector plus a "current": the wind's velocity minus the ship's velocity. Once we know that current, each leg is a river-crossing problem. We choose the heading so that the resultant points along the line, and then the out-and-back time fixes the distance.

Solution

Throughout, take x east and y north, and measure speeds in knots (nautical miles per hour), so that distances come out in nautical miles. The ship's velocity is $(-V, 0)$. If the air moves with velocity $\mathbf{w}$ and the plane's velocity relative to the air is $\mathbf{a}$ with $|\mathbf{a}| = kV$, then the plane's velocity relative to the ship is

$$\mathbf{a} + \mathbf{c}, \qquad \mathbf{c} = \mathbf{w} - (-V, 0) = \mathbf{w} + (V, 0).$$

The plane stays north-west of the ship, so $\mathbf{a} + \mathbf{c}$ must point along $\mathbf{u} = \frac{1}{\sqrt{2}}(-1, 1)$ on the way out and along $-\mathbf{u}$ on the way back.

With the wind: the outward flight

The wind blows *from* north *to* south, so $\mathbf{w} = (0, -V)$ and

$$\mathbf{c} = (V, -V) = -\sqrt{2}\, V\, \mathbf{u}.$$

This current has speed $\sqrt{2}\, V$ and points due south-east, exactly along the patrol line. So the pilot simply points the plane along the line. Heading out (north-west) she flies straight into the current; heading back she flies with it:

$$\text{outward speed} = (k - \sqrt{2})V, \qquad \text{return speed} = (k + \sqrt{2})V.$$

Here we need $k > \sqrt{2}$: otherwise the plane could never get away from the ship along the north-west line.

If the plane reaches a distance d_w from the ship and each out-and-back trip takes $\frac{1}{2}$ hour, then

$$\frac{d_w}{(k-\sqrt{2})V} + \frac{d_w}{(k+\sqrt{2})V} = \frac{1}{2}$$
$$\frac{d_w}{V} \cdot \frac{2k}{k^2-2} = \frac{1}{2},$$

so

$$d_w = \frac{(k^2-2)V}{4k}.$$

The outward flight therefore lasts

$$\frac{d_w}{(k-\sqrt{2})V} = \frac{k^2-2}{4k(k-\sqrt{2})} = \frac{k+\sqrt{2}}{4k} \text{ hours} = 15\left(1+\frac{\sqrt{2}}{k}\right) \text{ minutes.}$$

This is always more than 15 minutes, as it should be: the outward leg runs against the current.

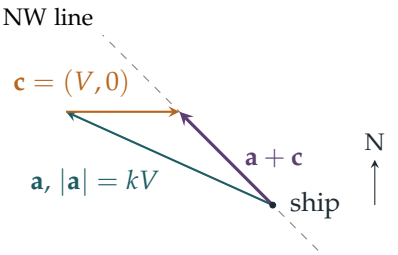
Ask yourself Why can't the plane just point north-west in calm air as well?

In calm air the current is no longer along the patrol line. If the pilot pointed the plane north-west, the eastward part of the current would carry it off the line. She has to aim slightly into the current, and that uses up some of her airspeed.

In calm air

Now $\mathbf{w} = \mathbf{0}$, so $\mathbf{c} = (V, 0)$: relative to the ship, the air streams east at V. Split $\mathbf{c}$ along the line, using $\mathbf{u}$, and across it, using $\mathbf{v} = \frac{1}{\sqrt{2}}(1,1)$:

$$\mathbf{c} = -\frac{V}{\sqrt{2}}\,\mathbf{u} + \frac{V}{\sqrt{2}}\,\mathbf{v}.$$



The resultant must have no $\mathbf{v}$-component, so $\mathbf{a}$ must have $\mathbf{v}$-component $-V/\sqrt{2}$. Since $|\mathbf{a}| = kV$, its component along the direction of travel is then

$$\sqrt{k^2V^2 - \tfrac{1}{2}V^2} = V\sqrt{k^2 - \tfrac{1}{2}}.$$

Along the line, the current contributes $-V/\sqrt{2}$ on the way out and $+V/\sqrt{2}$ on the way back (because $\mathbf{c}\cdot(-\mathbf{u}) = V/\sqrt{2}$). Write $A = V\sqrt{k^2-\frac{1}{2}}$ and $B = V/\sqrt{2}$. Then

$$\text{outward speed} = A - B, \qquad \text{return speed} = A + B.$$

The outward speed is positive because $A^2 - B^2 = V^2(k^2-1) > 0$.

With the same half-hour round trip,

$$\frac{d_c}{A-B} + \frac{d_c}{A+B} = \frac{2A\,d_c}{A^2-B^2} = \frac{1}{2} \quad\Longrightarrow\quad d_c = \frac{A^2-B^2}{4A} = \frac{V(k^2-1)}{4\sqrt{k^2-\frac{1}{2}}}.$$

The ratio

Dividing,

$$\frac{d_w}{d_c} = \frac{(k^2-2)V}{4k}\cdot\frac{4\sqrt{k^2-\frac{1}{2}}}{V(k^2-1)} = \frac{(k^2-2)\sqrt{k^2-\frac{1}{2}}}{k(k^2-1)}.$$

Since $\sqrt{k^2-\frac{1}{2}} = \frac{1}{2}\sqrt{4k^2-2}$,

$$\frac{d_w}{d_c} = \frac{k^2-2}{2k(k^2-1)}\sqrt{4k^2-2},$$

which is the required result.

Checks

- As $k \to \infty$ the wind and the ship hardly matter. The outward flight tends to 15 minutes, and $d_w/d_c \to 1$ (top and bottom of the ratio both grow like $2k^3$).
- As $k \to \sqrt{2}^{+}$ the windy outward speed falls to zero, so $d_w \to 0$ and the ratio $\to 0$, as the formula says.
- For $k = 2$ the outward flight takes $15(1+1/\sqrt{2}) \approx 25.6$ minutes, and $d_w/d_c = \frac{2}{12}\sqrt{14} \approx 0.624$. We confirmed this numerically by solving for the heading directly and by stepping the motion.

Common slips

- Getting the wind's direction backwards: a wind "from north to south" blows *towards* the south.
- Adding the ship's velocity instead of subtracting it. Relative to a westbound ship, still air drifts *east*.
- In calm air, pointing the plane north-west and using $kV \pm V/\sqrt{2}$ as the speeds. The cross-current has to be cancelled first.

Beyond STEP

This is a Galilean change of reference frame: velocities measured in frames moving uniformly relative to each other differ by a constant vector. The same bookkeeping underlies the Michelson–Morley "boat in a river" argument. There, the across-and-back trip and the along-and-back trip take different times in exactly the way they do here, and that is the classical calculation that special relativity overturned. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats relative motion and then relativity in this spirit.

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